Retrospective Sprint IX

Team 04

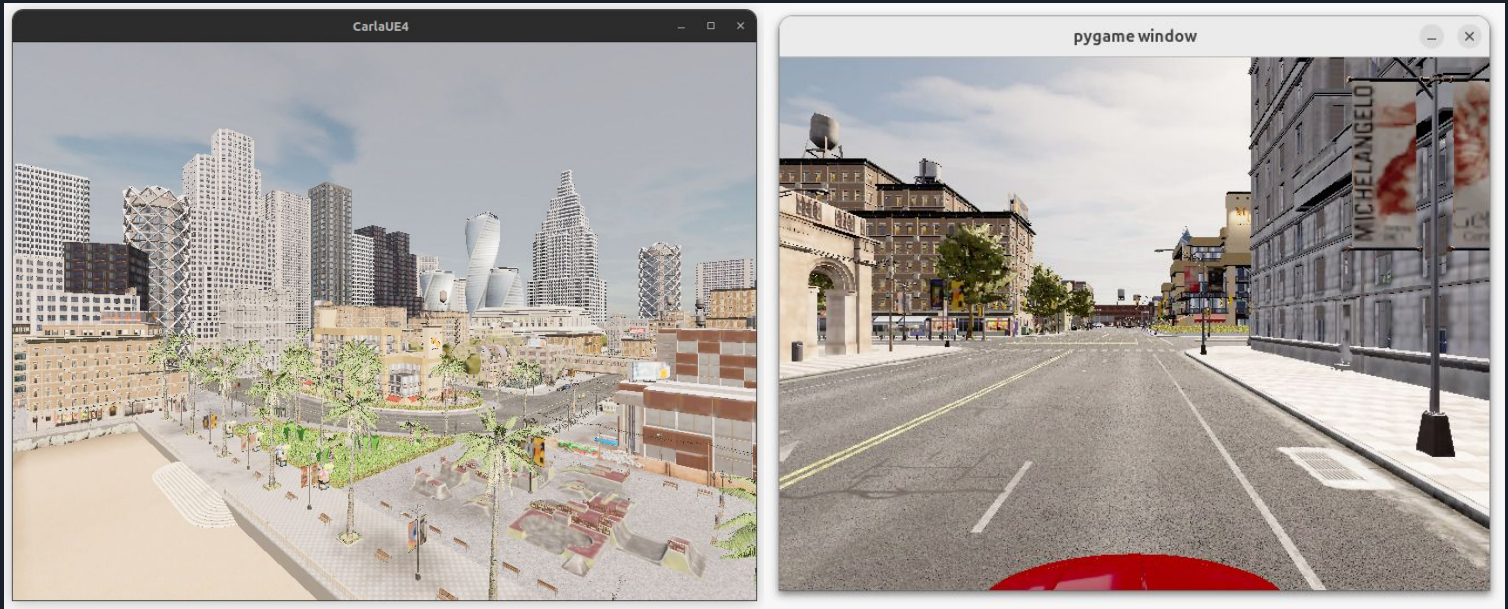
DrivaPi

Sprint IX Goal

Build a fully validated vehicle by finalizing **STM32 emergency logic** and **Qt telemetry**, establishing the **Camera/HAILO/CARLA** stack for ADAS, and enforcing strict **TSF traceability** through **real-code testing**.

CARLA simulator

Set up on a dedicated PC to run as a persistent simulation server, accessible over the network for model training



Training Method for Autonomous Navigation

IMITATION LEARNING

REINFORCEMENT LEARNING

DATASET AGGREGATION

Training Method for Autonomous Navigation

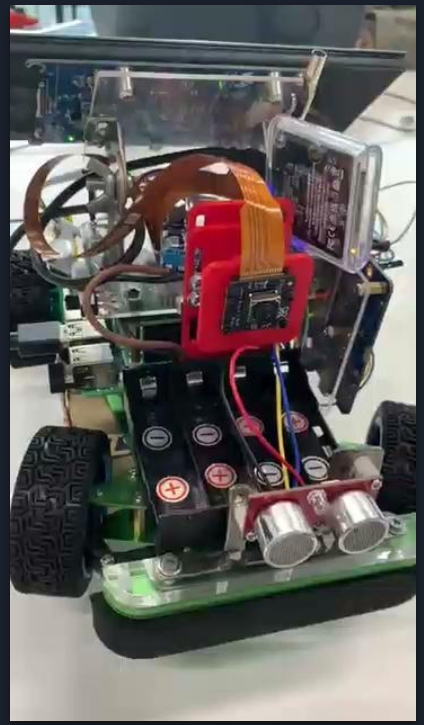
Criteria	Imitation Learning	Reinforcement Learning	Dataset Aggregation
Needs reward fn	No	Yes	No
Data Source	Human Demos	Self-exploration	Human + agent data
Distribution Shift	Severe	N/A	Largely Resolved
Performance Ceiling	Human-Level	Beyond Human	Near human-level
Training Cost	Low	Very High	Medium
Safety during Training	High	Low	Medium

Lane Keep Assist (LKA) research

Lane detection models for the HAILO NPU, the mathematical steering correction model, and a theoretical data flow from the vision pipeline through CAN bus to the STM32 control unit.

Camera and HAILO set-up

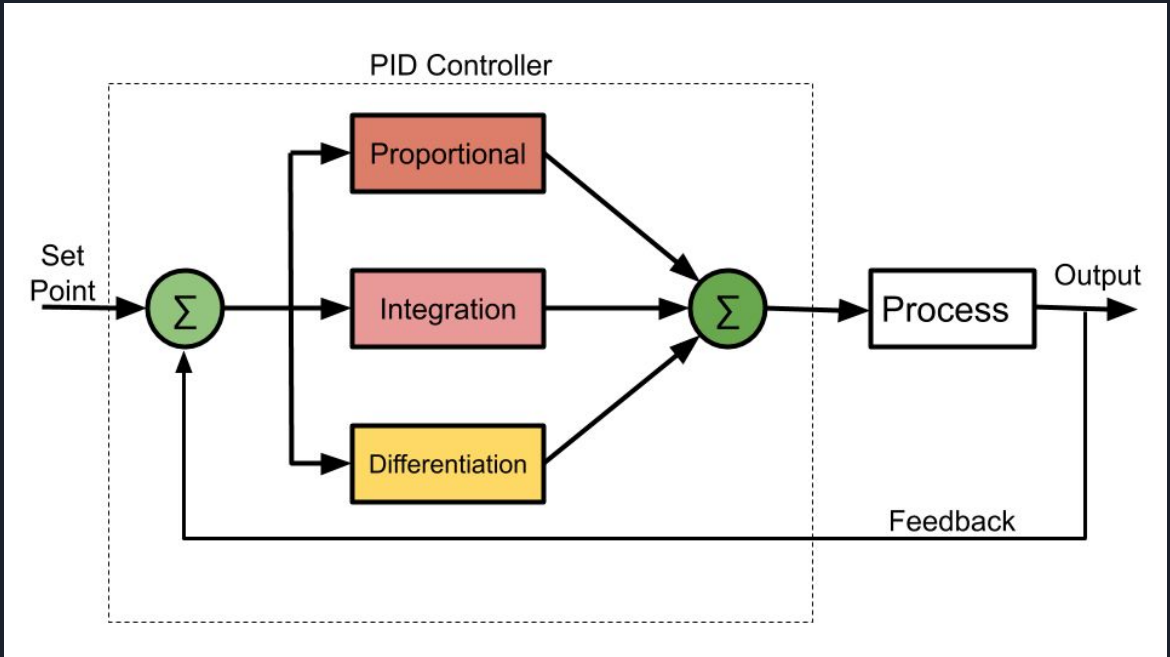
- Hailo-8 Configuration for AGL Trout Scarthgap working
- IMX708 NoIR Wide camera works on AGL Scarthgap
    - detected at boot and listed by rpicam-hello
- 3-part camera mount assembled and installed on the car



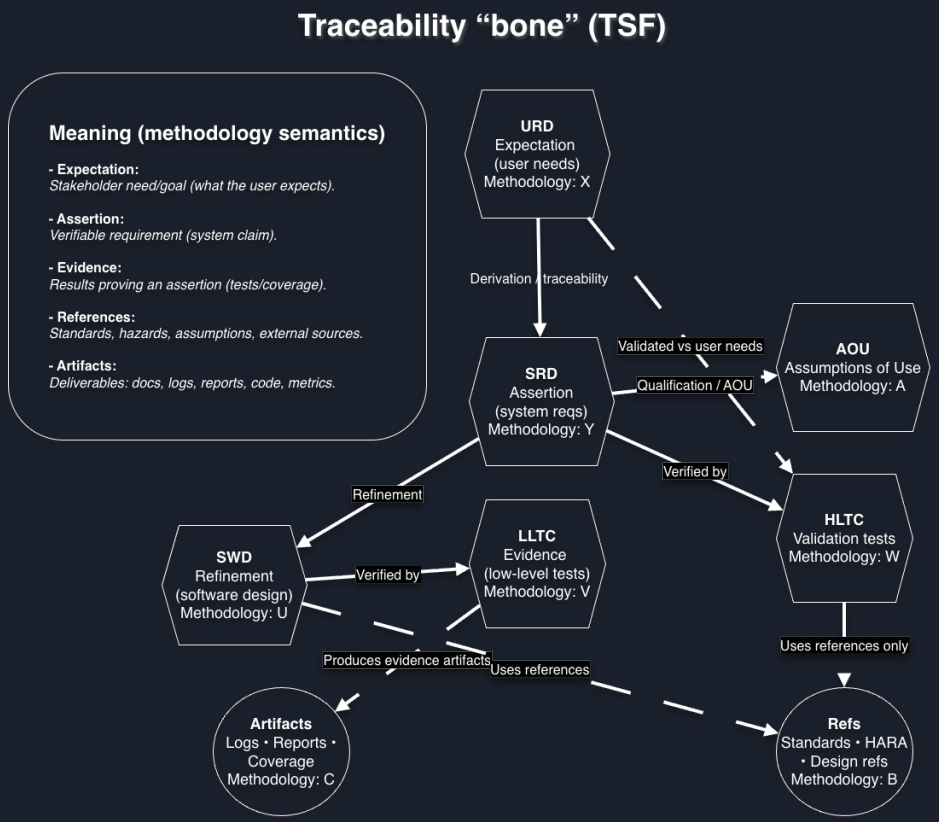
Camera set-up

PID - Proportional-Integral-Derivative

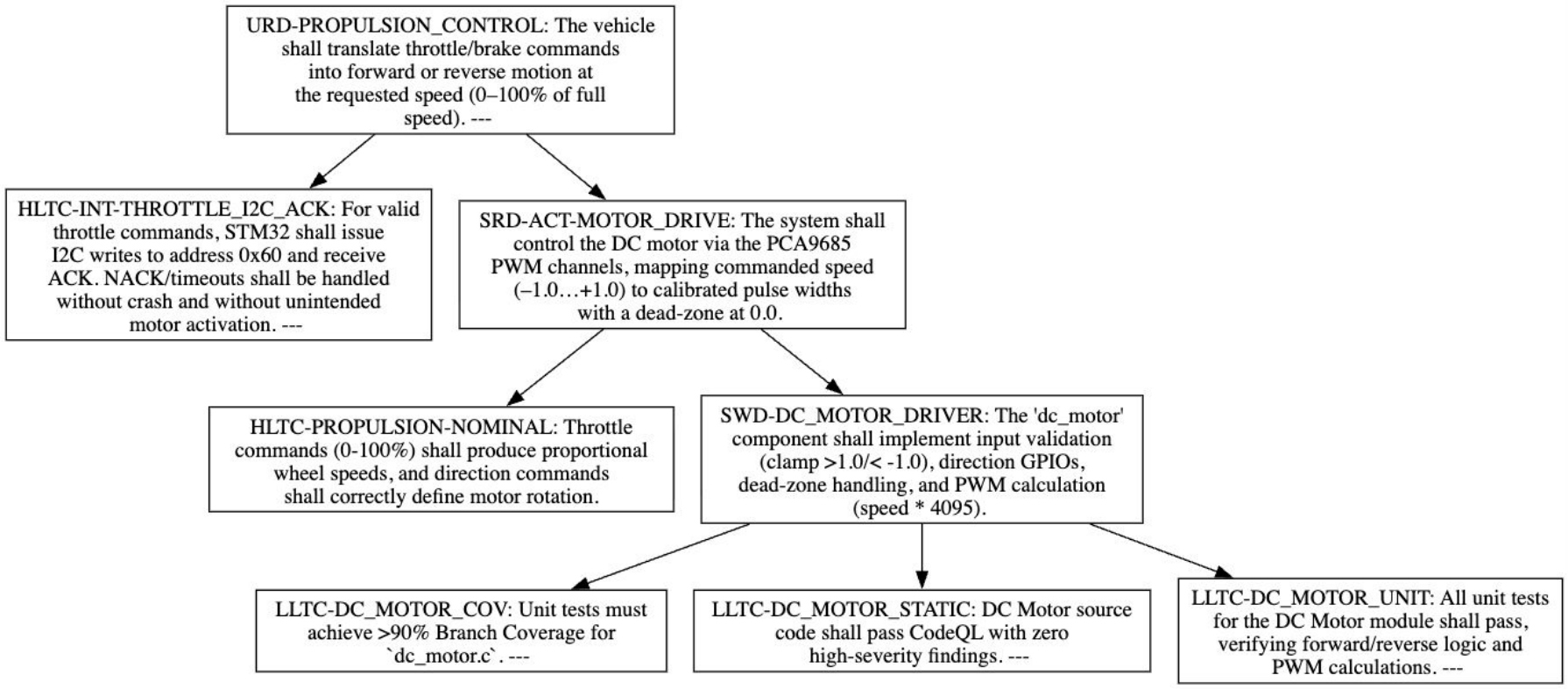
- PID control loops for smooth/accurate DC motor speed control (research first, then implement + tune)



TSF Structure



H-02 → SG-02 → URD


URD-PROPULSION_CONTROL: The vehicle shall translate throttle/brake commands into forward or reverse motion at the requested speed (0–100% of full speed). ---
HLTC-INT-THROTTLE_I2C_ACK: For valid throttle commands, STM32 shall issue I2C writes to address 0x60 and receive ACK. NACK/timeouts shall be handled without crash and without unintended motor activation. ---
SRD-ACT-MOTOR_DRIVE: The system shall control the DC motor via the PCA9685 PWM channels, mapping commanded speed (–1.0…+1.0) to calibrated pulse widths with a dead-zone at 0.0.
HLTC-PROPULSION-NOMINAL: Throttle commands (0-100%) shall produce proportional wheel speeds, and direction commands shall correctly define motor rotation.
SWD-DC_MOTOR_DRIVER: The 'dc_motor' component shall implement input validation (clamp >1.0/< -1.0), direction GPIOs, dead-zone handling, and PWM calculation (speed * 4095).
LLTC-DC_MOTOR_COV: Unit tests must achieve >90% Branch Coverage for `dc_motor.c`. ---
LLTC-DC_MOTOR_STATIC: DC Motor source code shall pass CodeQL with zero high-severity findings. ---
LLTC-DC_MOTOR_UNIT: All unit tests for the DC Motor module shall pass, verifying forward/reverse logic and PWM calculations. ---

COVERAGE

UNIT

```
Team04_DrivaPi ❯ tree
verification-artifacts
verification-artifacts
├── coverage
│   ├── coverage.info
│   ├── coverage.xml
│   ├── dc-motor.xml
│   ├── servo-motor.xml
│   └── speed-sensor.xml
└── tests
    ├── dc-motor.xml
    ├── junit_results.xml
    ├── servo-motor.xml
    ├── speed-sensor.xml
    └── summary.json

3 directories, 10 files
```

STATIC ANALYSIS

cpp.sarif

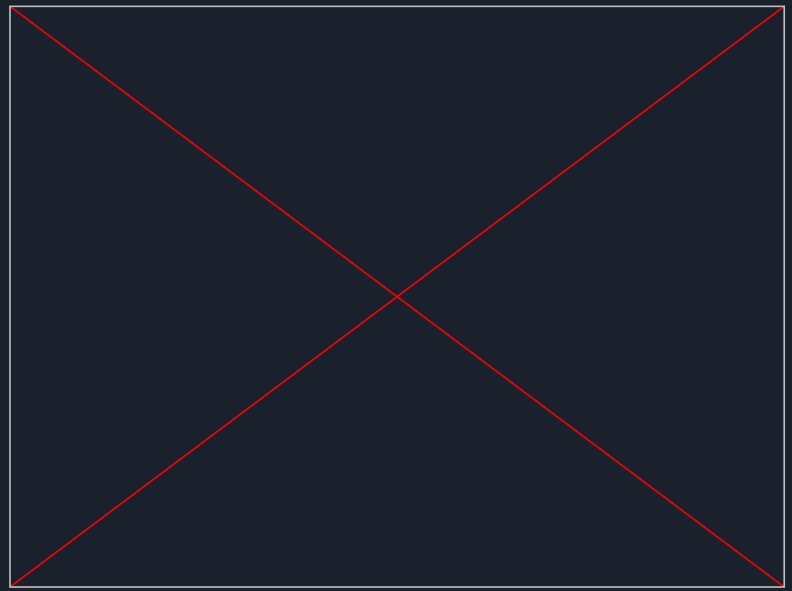
CUSTOM VALIDATOR

Scoring

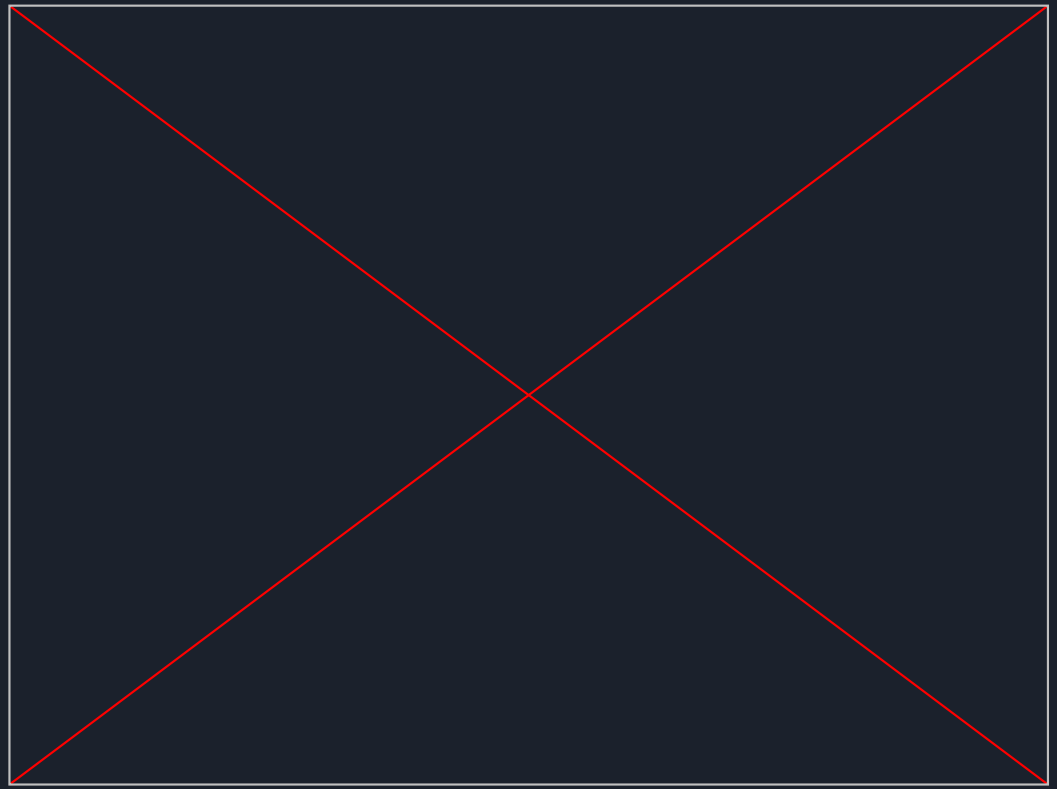
URD-COMM_INTEGRITY = 0.0; Missing
URD-DETERMINISTIC_SCHED = 0.0; Missing
URD-DRIVER_DASHBOARD = 0.0; Missing
URD-EMERGENCY_STOP = 0.0; Missing
URD-FAULT_FAILSAFE = 0.0; Missing
URD-HMI_COMMUNICATION = 0.0; Missing
URD-LANE_DETECTION = 0.0; Missing
URD-OBSTACLE_AVOIDANCE = 0.0; Missing
URD-PROPULSION_CONTROL = 0.83333; Derived from supporting Statements
URD-RESOURCES_MONITOR = 0.0; Missing
URD-SPEED_MEASUREMENT = 0.55556; Derived from supporting Statements
URD-STARTUP_CALIBRATION = 0.0; Missing
URD-STEERING_CONTROL = 0.55556; Derived from supporting Statements

System/Functional Tests

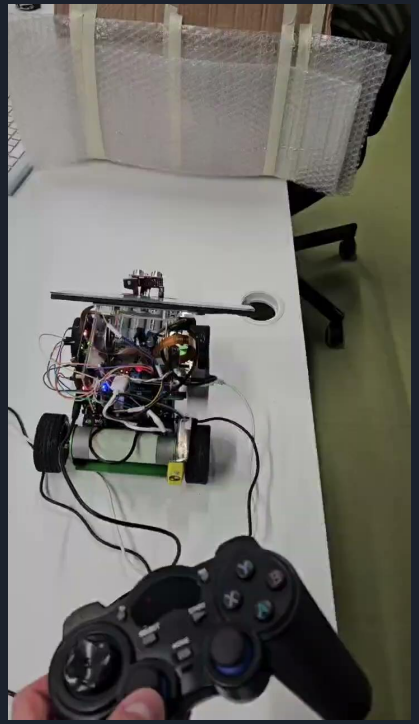
https://github.com/SEAME-pt/Team04_DrivaPi/blob/78dbca8fdf03bd95a7d6665fa2e4cfb889929861/docs/tests/speed_accuracy_system_test.md



QT improvement

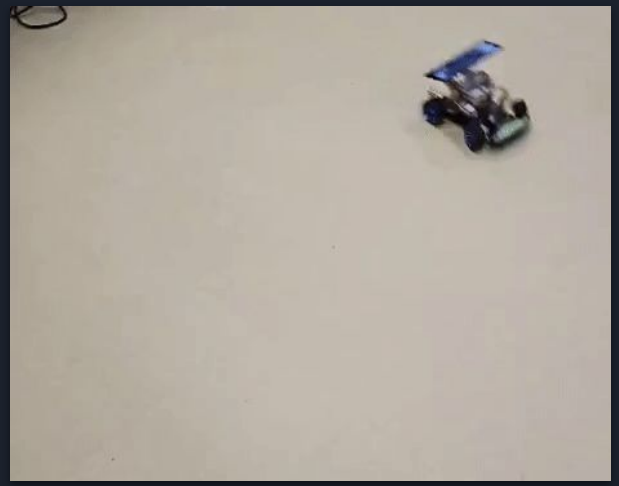


Emergency braking



EXTRAS

Logotypes + Instructions of usage

AI Instructions improvement

Thank you!